



5.8.6 Wrap-Up: Ticket Out the Door

1. Solve the system of equations using substitution.

$$\begin{cases} 6x - 5y = -8 \\ y = -\frac{1}{2}x + 5 \end{cases}$$

Unit 5, Lesson 9: Solving Systems of Equations by Elimination



Benchmark

MA.912.AR.9.1 Given a mathematical or real-world context, write and solve a system of two-variable linear equations algebraically or graphically.



Learning Target

- ☐ I can solve systems of equations by elimination.



5.9.1 Warm-Up: Goals and Aspirations

Nilo Cruz is a Cuban American playwright. Born in Matanzas, Cuba in 1960, his family immigrated to Miami, Florida in 1970. He first studied theater at Miami-Dade Community College before moving to New York City to begin his career. He later earned his Master's in Fine Arts from Brown University in 1994. Cruz served as the playwright-in-residence for the New Theater in Coral Gables, Florida in 2001. While there, he wrote *Anna in the Tropics*, a play he later won a Pulitzer Prize for in 2003. Nilo Cruz was the first Latino to win the Pulitzer Prize for Drama.



1. If you win a prize for a major accomplishment in the future, what would you want to be recognized for?



5.9.2 Guided Instruction: Solving Systems of Linear Equations Using Elimination

In addition to solving systems of equations by graphing and using substitution, systems of equations can also be solved algebraically using **elimination**.

Solving a system by elimination means to *eliminate* one of the variables by adding or subtracting the system together.

1. Consider the system of equations $\begin{cases} 2x + 4y = 13 \\ -2x + 4y = 11 \end{cases}$

- a. Add the two equations together.

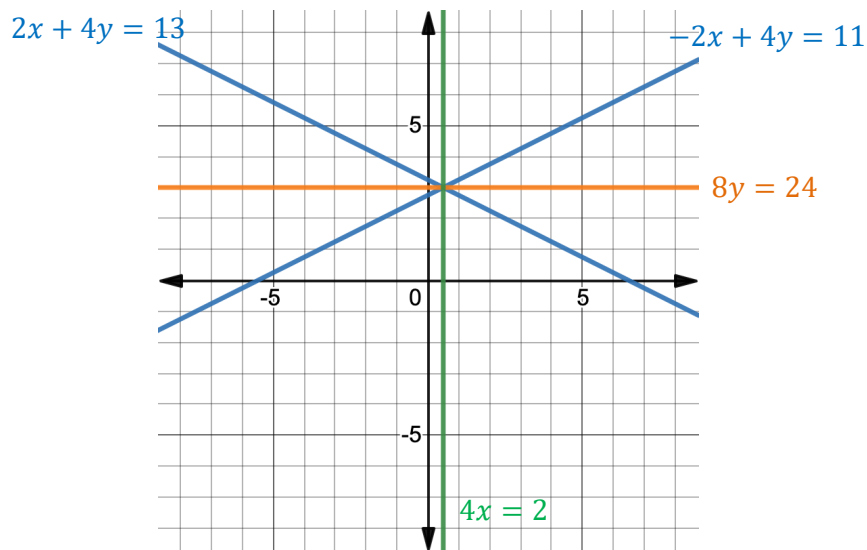
$$\begin{array}{r} 2x + 4y = 13 \\ +(-2x + 4y = 11) \\ \hline \end{array}$$

- b. Subtract one equation from the other.

$$\begin{array}{r} 2x + 4y = 13 \\ -(-2x + 4y = 11) \\ \hline \end{array}$$

- c. What do you notice about the resulting equations in Parts A and B?

- d. The graphs of the original equations in the system and the resulting equations from Parts A and B are shown on the same coordinate grid.



What do you notice?

- e. Use the resulting equation from Part A to solve the original system of equations algebraically.

- f. Verify your solution using the graph.



When equations in a system are added or subtracted, the resulting equation shares the solution of the original system.

2. Use the system of equations $\begin{cases} 5a - b = 9 \\ 2.5a + b = 6 \end{cases}$ to complete the following.

- a. Complete the statement.

- ☐ Adding
☐ Subtracting

the original equations in this

system would result in eliminating one of the

variables,

- ☐ a .
☐ b .

- b. Perform the operation selected in Part A.

$$\begin{array}{r} 5a - b = 9 \\ \underline{-(2.5a + b = 6)} \end{array}$$

- c. Solve the system using your equation from Part B.

$$a = \underline{\hspace{2cm}}$$

$$b = \underline{\hspace{2cm}}$$

- d. Riley wondered if it is possible to solve the system by eliminating the a -terms instead of the b -terms first. He suggested multiplying the second equation by 2 and then subtracting it from the first equation. The first steps Riley suggests are shown. Complete Riley's work and use the results to solve the system.

$$2(2.5a + b = 6)$$

$$\boxed{}a + \boxed{}b = \boxed{}$$

$$\begin{array}{r} 5a - b = 9 \\ -() \\ \hline \end{array}$$

$$a = \underline{\hspace{2cm}}$$

$$b = \underline{\hspace{2cm}}$$



When solving a system of equations using **elimination**, one or both of the equations can be multiplied by a factor before adding or subtracting the equations. The resulting equation shares the same solution with the original system.



5.9.3 Guided Practice: Solving Systems of Linear Equations Using Elimination

1. Use the system of equations to complete the following.

$$\begin{cases} 3m + 4k = 24 \\ 2m - 8k = -12 \end{cases}$$

- a. Solve the system using elimination.

$$m = \underline{\hspace{2cm}}$$

$$k = \underline{\hspace{2cm}}$$

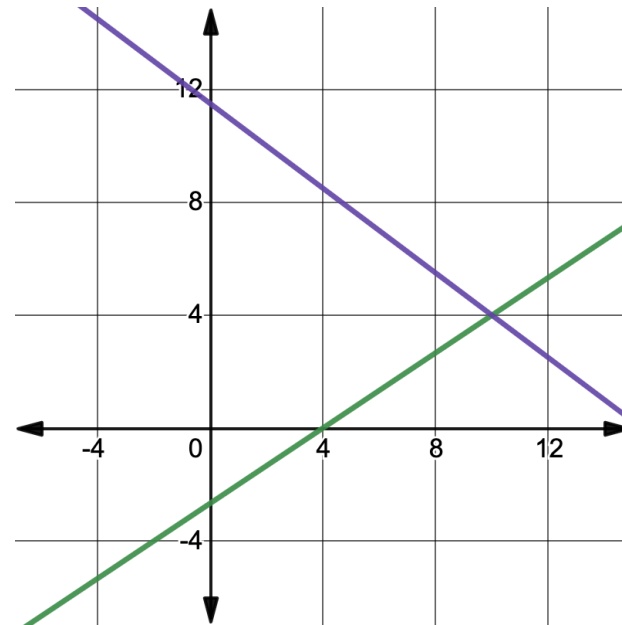
- b. Verify the solution algebraically using substitution.

2. Sometimes in order to solve a system by elimination, both equations must first be multiplied by different factors to eliminate one variable when adding or subtracting.

a. Discuss with your partner what factors could be multiplied by either equation in the system $\begin{cases} 2x - 3y = 8 \\ 3x + 4y = 46 \end{cases}$ to solve by elimination.

b. Solve the system using elimination.

c. The graph of the system is shown.



Discuss with your partner whether or not the graph verifies the solution found in Part B. If it does not, revisit Part B to find any errors.



5.9.4 Your Turn!

1. Solve the system of equations using elimination.

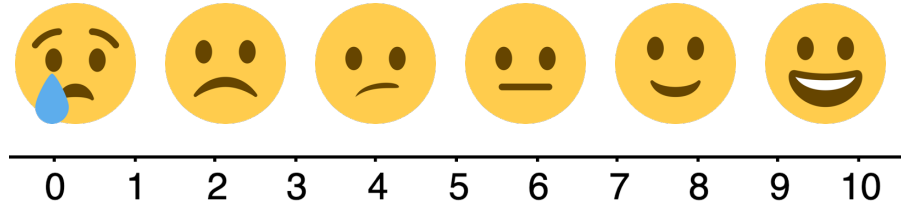
$$\begin{cases} 4x - y = -10 \\ -12x + 7y = 22 \end{cases}$$

2. Verify the solution algebraically.



5.9.5 Wrap-Up: Reflection

1. Use the emoji scale to indicate your level of confidence with solving systems of equations using elimination.



2. Explain why you chose that level of confidence.